

mmWave Channel Modelling and Performance Indicators

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Abstract

The domain of wireless communication has expanded manifold in recent years. Applications such self driving vehicles and 5G mobile communication demand an extensive as well as efficient utilization of bandwidth. Such ambitious goals along with the ever increasing capacity demand can only be fulfilled by moving upward in the spectrum. This calls for a comprehensive and wholesome channel modelling which would take into account the major factors responsible for degradation of signals over the wireless medium. The project is based on channel sounding measurements in a vehicular/outdoor scenario centered around a specific license free band of 60GHz.

1 Motivation

In most of the cases, the physical properties of the channel itself is responsible for introducing undesirable changes into the signals while in transit to the receiver in a wireless communication system. As mentioned in [1], the 60 GHz are extremely sensitive to factors such as droplets of rain, fog and gas. Electromagnetic diffraction as well as reflection of the signals from terrestrial objects such as buildings and trees generate several copies of the same signal, known as multi-path signals. These are essentially disparate in terms of phase shift, amplitude and delays. Furthermore destructive interference of such signals at certain points in the channel results into a complete loss of signal power and a consequent drop in signal-to-noise ratio (SNR). In addition to the problem of multi-path signals, the phenomenon of Doppler effect caused due to relative motion between the transmitter and the receiver renders the signal parameters as time varying. As a result, without a proper channel state information (CSI) at the receiver end, one cannot opt for coherent demodulation technique. Non coherent demodulation techniques such as differential demodulation comes at an expense of 3-4 dB loss in signal-to-noise ratio when compared with its counterpart[2]. Hence the only viable solution at hand is to accurately model the channel to ensure the communication link is above a satisfactory level.

2 Project Description

Existing communication systems employ at large multiple antennas with an obvious advantage of enhanced channel capacity and quality-of-service (QoS). However for exploiting the channel efficiently, one needs to be aware of time variant characteristics of the channel. Several multipath parameters including time delay, Doppler shift, direction-of-arrival (DoA), direction-of-departure (DoD) aid to the precise characterisation of the time varying channel. The entire project has three objectives:

- Preparation of experimental setup for wireless channel measurement at 60GHz band.
- Extensive analysis of small scale and large scale fading parameters.
- Development of analytical channel model.

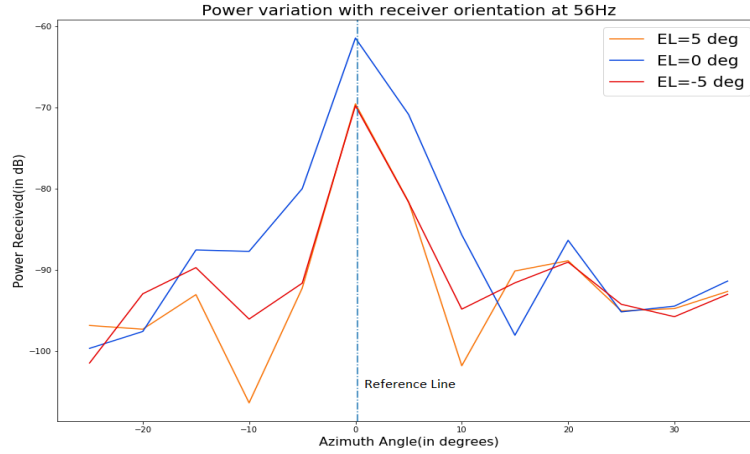


Figure 1: Variation with received power with receiver orientation

The data collected from the experimental setup can be used to extract the channel transfer function in the frequency domain. Furthermore a windowing operation followed by an IFFT operation would yield us the channel impulse response (CIR). This will enable us to create the power delay profile along with estimation of certain small scale fading parameters such as RMS delay and Doppler spread. The power at the receiver end clearly exhibits that the maximum signal-to-noise ratio results when the receiver is in direct line-of-sight with the transmitter. Such a direction of maximum signal strength as shown in Figure 1 (elevation angle = 0° , azimuth angle = 0°) is taken as the reference. Measurements in all other directions are made relative to it.

3 My Contribution

My task in the project would be to accurately analyze the extracted parameters to develop an analytical model, which would not only take the fading factors into account but also would serve as a reference for actual propagation in hot and humid climatic conditions. This would involve working with multiple datasets in a Matlab based environment. As the project aims to measure, analyze and model radio wave propagation in the 60GHz cw-band, the resultant model will serve as a indispensable resource for industries and various research groups alike. Additionally, this would reinforce the existing propagation standards set by various standardization bodies such as IEEE and will lead to subsequent development of 5G technologies.

4 References

References

- [1] Hitesh Singh, Ramjee Prasad, and Boncho Bonev. The studies of millimeter waves at **60 GHz** in outdoor environments for imt applications: A state of art. *Wireless Personal Communications*, 100(2):463–474, 2018.
- [2] YLJC Li, Leonard J Cimini, and Nelson R Sollenberger. Robust channel estimation for ofdm systems with rapid dispersive **fading channels**. *IEEE Transactions on communications*, 46(7):902–915, 1998.